

Mrs Cooper  
Mrs Greenberg  
Mrs Kerr  
Mrs Millar  
Mrs Squires

Mr Feng  
Ms Hoang  
Ms Lau  
Ms Peng  
Mrs Ward

Name: .....

Teacher: .....



*Pymble Ladies' College*

**YEAR 11**

**MATHEMATICS ADVANCED**

**ASSESSMENT TASK 1**

**Term 1  
2023**

**General Instructions**

- Working time 55 minutes
- Write using black or blue pen.
- Erasable pens are not to be used.
- Approved calculators may be used.
- Diagrams are not drawn to scale.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or untidy work.
- Reference sheets are provided.

|                     |            |
|---------------------|------------|
| <b>Mark</b>         | <b>/42</b> |
| <b>Rank</b>         | <b>/</b>   |
| <b>Highest Mark</b> | <b>/42</b> |

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# Mathematics

## Advanced

Name: .....

Teacher's Name: .....

### Section I – Multiple Choice Answer Sheet

Select the alternative (A), (B), (C) or (D) that best answers the question. Fill in the response oval completely.

**Sample:**      $2 + 4 =$      (A) 2     (B) 6     (C) 8     (D) 9  
                                         A ☐     B ☒     C ☐     D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒     B ☒     C ☐     D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒     B ☒     C ☐     D ☐  
                                         *correct*  
                                         ↙

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### Assessment Task 1

1     A ☐     B ☐     C ☐     D ☐

2     A ☐     B ☐     C ☐     D ☐

3     A ☐     B ☐     C ☐     D ☐

4     A ☐     B ☐     C ☐     D ☐

5     A ☐     B ☐     C ☐     D ☐

6     A ☐     B ☐     C ☐     D ☐

7     A ☐     B ☐     C ☐     D ☐

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## SECTION I

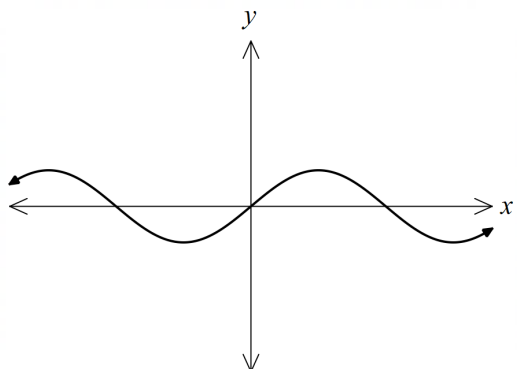
(7 Marks)

Answer using the Multiple Choice Answer Sheet provided.

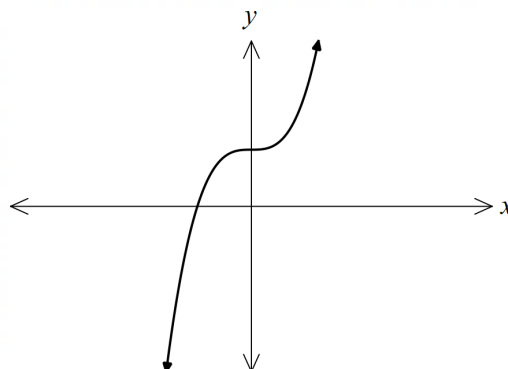
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1 Which graph shows the graph of an odd function?

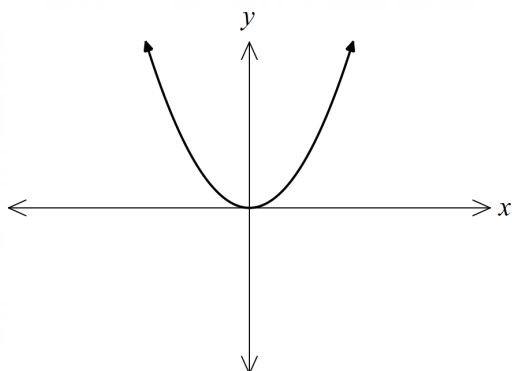
(A)



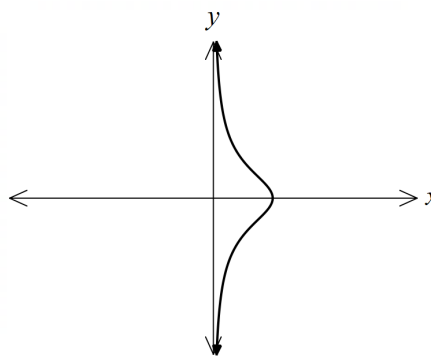
(B)



(C)



(D)



2 Which inequality defines the domain of the function  $f(x) = \frac{1}{\sqrt{x+3}}$ ?

(A)  $x > -3$

(B)  $x \geq -3$

(C)  $x < -3$

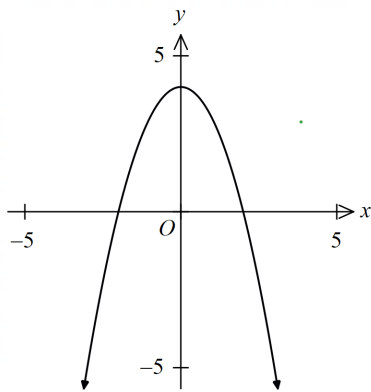
(D)  $x \leq -3$

3 Which of the following is true for the equation  $3x^2 - x - 2 = 0$ ?

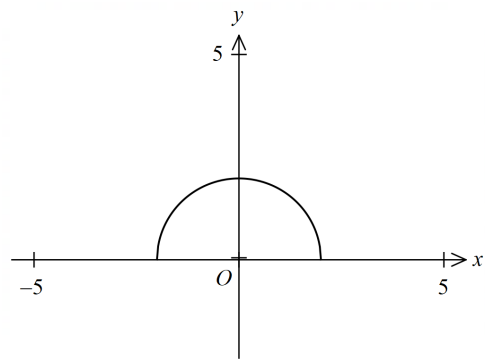
- (A) No real roots
- (B) One real root
- (C) Two real distinct roots that are rational
- (D) Two real distinct roots that are irrational

4 Which of the following functions is one-to-one?

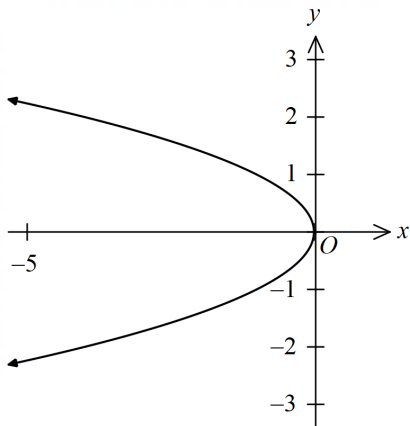
(A)



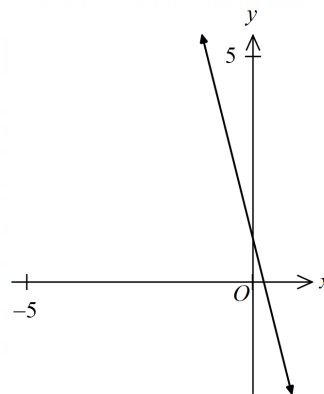
(B)



(C)



(D)



5 Which of the following is NOT equivalent to  $\sqrt{24}$ ?

(A)  $2\sqrt{6}$

(B)  $\frac{1}{2}\sqrt{48}$

(C)  $\sqrt{12} \times \sqrt{2}$

(D)  $2\sqrt{3} \times \sqrt{2}$

6 The function  $f(x) = \frac{x^2}{1-x^3}$  is which of the following:

(A) even

(B) odd

(C) neither

(D) cannot be determined

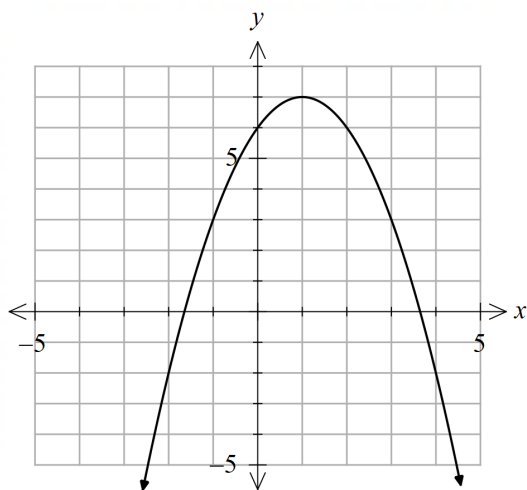
7 For the graph of the quadratic function below, what is the inequality that satisfies  $f(x) > 3$ ?

(A)  $(-\infty, -1] \cup [3, \infty)$

(B)  $(-\infty, -1) \cup (3, \infty)$

(C)  $[-1, 3]$

(D)  $(-1, 3)$



**End of Section I**

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**SECTION II**

(35 Marks)

Attempt all questions

Answer these questions in the spaces provided

Your responses should include relevant mathematical reasoning and/or calculations.

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**Question 8 (4 marks)****Marks**

(a) Solve for  $m$   $\frac{a^5 \sqrt{a}}{a^{-2}} = a^m$

**2**

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(b) Express  $y = 2x^2 - 12x + 11$  in the form  $y = a(x - h)^2 + k$ .

**2**

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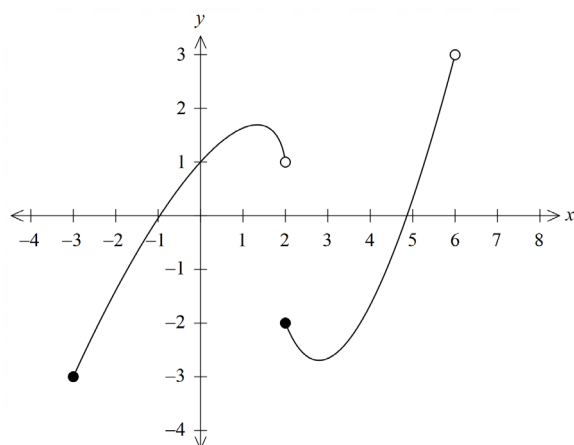
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**Question 9 (4 marks)****Marks**

- (a) State the domain and range of the following functions.

**2**

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- (b)  $y = 1 - \sqrt{1 - x}$

**2**

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**Question 10 (6 marks)****Marks**

- (a) A function is defined as  $f(x) = x^3 - 2x$ .

**2**

Show this function is either even, odd or neither.

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- (b) The function  $y = f(x)$  is defined as follows:

**2**

$$f(x) = \begin{cases} x^2 & \text{for } x \leq 2 \\ 2x + 1, & \text{for } x > 2 \end{cases}$$

Evaluate  $f(-1) + 2f(2) - f(3)$ .

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- (c) If  $g(x) = x^2 + 3x + 5$ , show that  $g(a+1) = g(a) + 2(a+2)$ .

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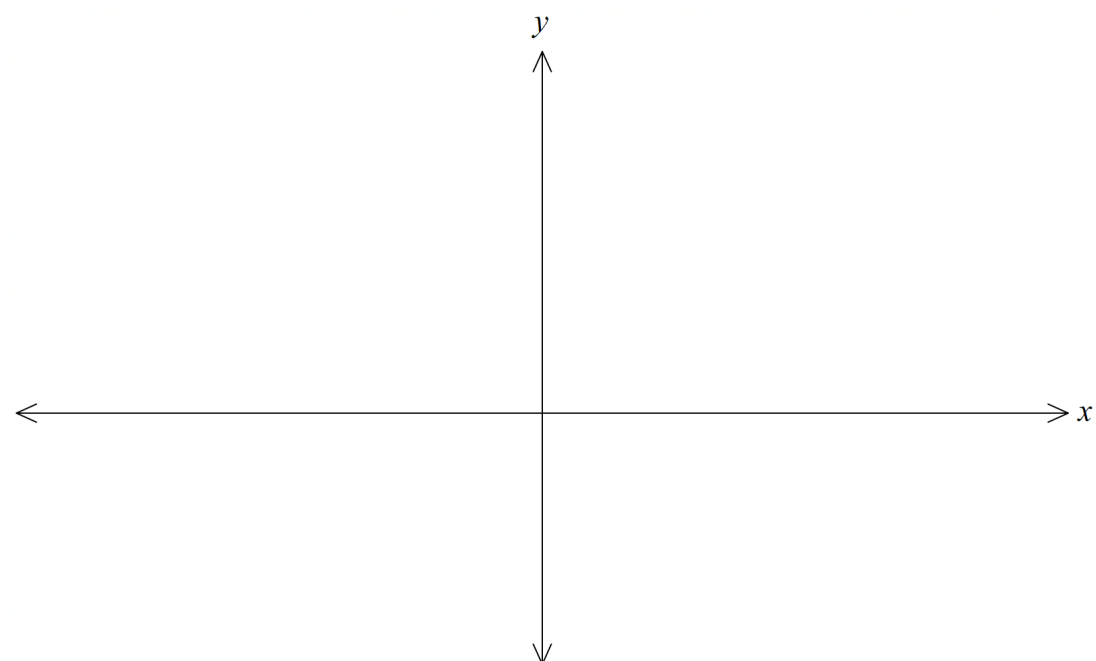
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**Question 11** (6 marks)**Marks**

- (a) (i) On the axis provided, sketch the graph  $y = x(x-4)^2$ , showing all intercepts.

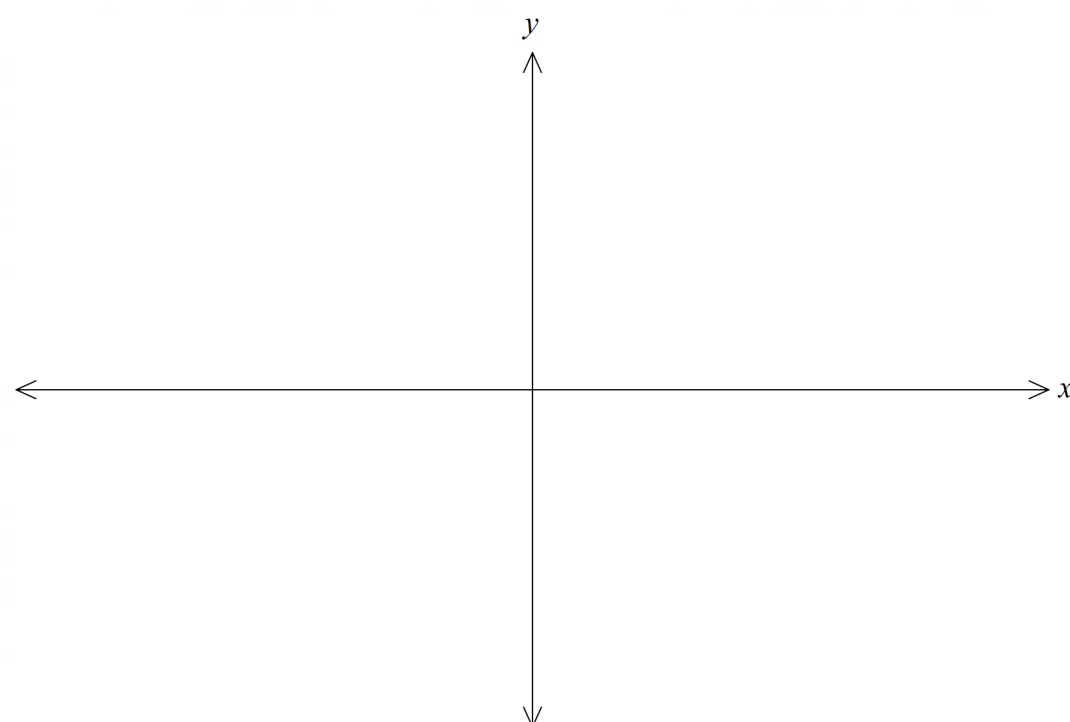
**2**

- (ii) Hence, find the values of  $x$  for which  $x(x-4)^2 > 0$ .

**2**

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- (iii) On the number plane below, sketch  $y = f(-x)$  labelling all intercepts.

**2**

**Question 12 (7 marks)****Marks**

- (a) Given that  $9x^3 + 3x^2 - 12x - 4 = (3x^2 - 4)(3x + 1)$ , hence, or otherwise, solve  $18x^4 + 6x^3 - 24x^2 - 8x = 0$ .

**2**

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- (b) For what values of  $m$  does the equation  $6x^2 - 12mx - 7m + 3 = 0$  have equal roots.

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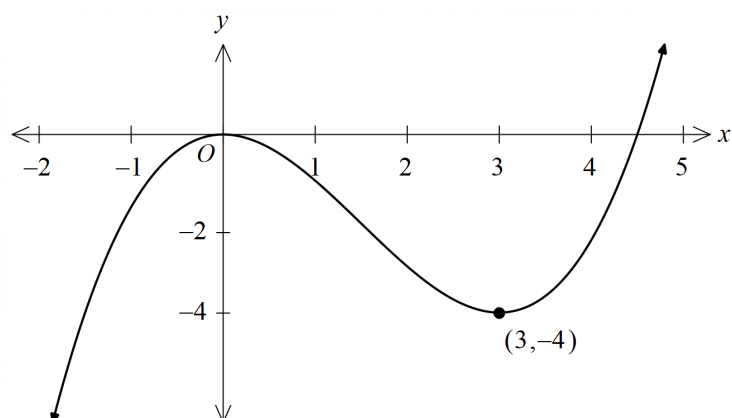
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**Question 12 continues on next page**

Question 12 continued.

(c) The graph of  $y = f(x)$  is sketched below.



For what values of  $k$  does the equation  $f(x) = k$  have one solution?

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**Question 13** (4 marks)

**Marks**

Let  $f(x) = -x^2 + x + 4$  and  $g(x) = x^2 - 2$ .

(a) Find  $g(f(3))$ .

**2**

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(b) Express  $f(g(x))$  in the form  $ax^4 + bx^2 + c$ , where  $a$ ,  $b$  and  $c$  are non-zero integers.

**2**

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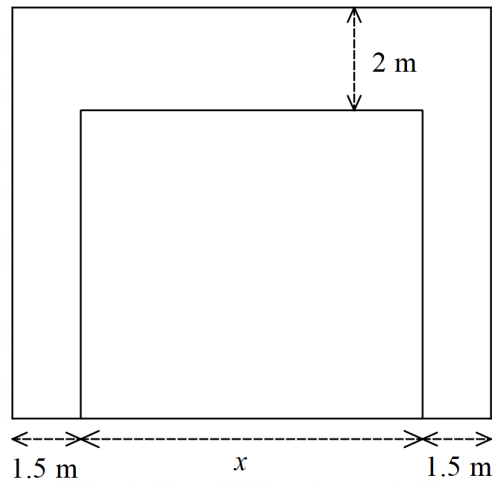
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**Question 14** (4 marks)

**Marks**

A square garage door of length  $x$  metres is surrounded on three sides, by brickwork. The brick work is 1.5 metres wide on each side of the door and has a height of 2 metres above the door, as shown below.



NOT TO SCALE

- (a) Find an expression for the area of the brickwork.

2

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- (b) To meet building regulations, the area of the brick wall must be exactly twice the area of the garage door.

2

Find the value(s) of  $x$  that satisfy the building regulations.

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Mathematics Advanced  
Mathematics Extension 1  
Mathematics Extension 2

REFERENCE SHEET

**Measurement**

**Length**

$$l = \frac{\theta}{360} \times 2\pi r$$

**Area**

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

**Surface area**

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

**Volume**

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

**Functions**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For  $ax^3 + bx^2 + cx + d = 0$ :

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

**Relations**

$$(x - h)^2 + (y - k)^2 = r^2$$

**Financial Mathematics**

$$A = P(1 + r)^n$$

**Sequences and series**

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

**Logarithmic and Exponential Functions**

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

## Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

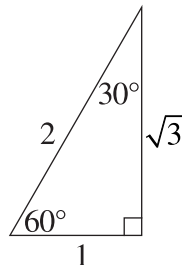
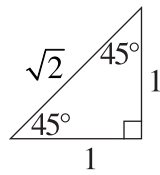
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



### Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

### Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1 + t^2}$$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

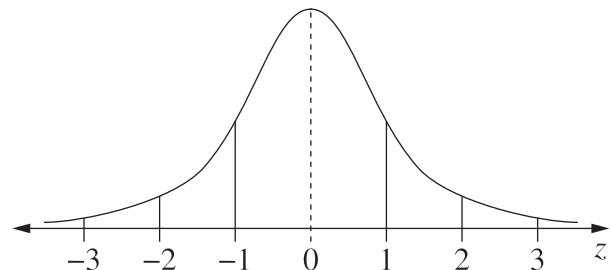
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

## Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score  
less than  $Q_1 - 1.5 \times IQR$   
or  
more than  $Q_3 + 1.5 \times IQR$

### Normal distribution



- approximately 68% of scores have  $z$ -scores between  $-1$  and  $1$
- approximately 95% of scores have  $z$ -scores between  $-2$  and  $2$
- approximately 99.7% of scores have  $z$ -scores between  $-3$  and  $3$

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

### Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

### Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

### Binomial distribution

$$P(X = r) = {}^nC_r p^r (1 - p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1 - p)$$

## Differential Calculus

### Function

### Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

## Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where  $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \left\{ f(a) + f(b) + 2[f(x_1) + \cdots + f(x_{n-1})] \right\}$$

where  $a = x_0$  and  $b = x_n$

## Combinatorics

$${}_nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}_nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

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## Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

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## Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

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## Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

# Mathematics

## Advanced

Name: .....

Teacher's Name: .....

### Section I – Multiple Choice Answer Sheet

Select the alternative (A), (B), (C) or (D) that best answers the question. Fill in the response oval completely.

Sample:  $2 + 4 =$  (A) 2 (B) 6 (C) 8 (D) 9

A ☒ B ☐ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐  
correct  
↑

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### Assessment Task 1

1 A ☒ B ☐ C ☐ D ☐

2 A ☒ B ☐ C ☐ D ☐

3 A ☐ B ☐ C ☒ D ☐

4 A ☐ B ☐ C ☐ D ☒

5 A ☐ B ☒ C ☐ D ☐

6 A ☐ B ☐ C ☒ D ☒

7 A ☐ B ☐ C ☐ D ☒

## SECTION II

(35 Marks)

Attempt all questions

Answer these questions in the spaces provided

Your responses should include relevant mathematical reasoning and/or calculations.

### Question 8 (4 marks)

Marks

(a) Solve for  $m$   $\frac{a^5 \sqrt{a}}{a^{-2}} = a^m$

2

$$\frac{a^{11/2}}{a^{-2}} = a^{15/2}$$

$$= a^{11/2 - (-2)} \quad \therefore m = 15/2$$

|   |         |
|---|---------|
| 2 | Correct |
| 1 | error   |

(b) Express  $y = 2x^2 - 12x + 11$  in the form  $y = a(x-h)^2 + k$ .

2

$$y = 2(x^2 - 6x + 9) + 11 - 18$$

$$= 2(x-3)^2 - 7$$

|   |                                       |
|---|---------------------------------------|
| 2 | Correct                               |
| 1 | Correctly<br>completing the<br>square |

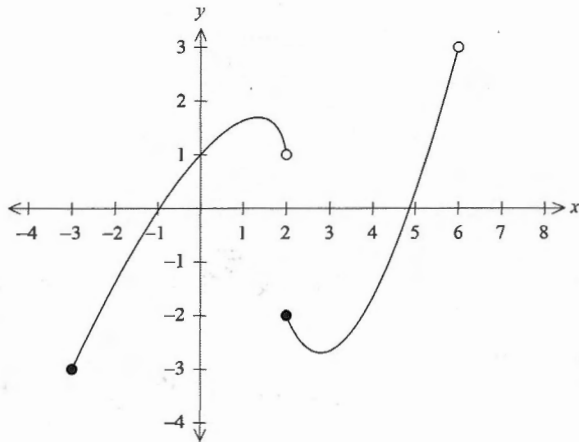
having  $(x-3)^2$

Question 9 (4 marks)

Marks

(a) State the domain and range of the following functions.

2



① max if no  $x \in$   
 $y \in$   
and both D and R  
correct

|                          |   |              |
|--------------------------|---|--------------|
| Domain : $x \in [-3, 6)$ | 2 | Both correct |
| Range : $y \in [-3, 3)$  | 1 | one correct  |

(b)  $y = 1 - \sqrt{1-x}$

2

|                                      |   |              |
|--------------------------------------|---|--------------|
| Domain : $1-x \geq 0 \quad x \leq 1$ | 2 | Both correct |
| Range : $y \leq 1$                   | 1 | one correct  |

① max if no  $x \in$   
 $y \in$   
and both D and R  
correct.

Question 10 (6 marks)

Marks

- (a) A function is defined as  $f(x) = x^3 - 2x$ .

2

Show this function is either even, odd or neither.

$$\begin{aligned}
 f(a) &= a^3 - 2a \\
 f(-a) &= (-a)^3 - 2(-a) \\
 &= -a^3 + 2a \\
 &= -(a^3 - 2a) \\
 &= -f(a) \\
 \therefore \text{function is odd}
 \end{aligned}$$

|   |                             |
|---|-----------------------------|
| 2 | Correct                     |
| 1 | Subs $x = -a$ or equivalent |

- (b) The function  $y = f(x)$  is defined as follows:

2

$$f(x) = \begin{cases} x^2 & \text{for } x \leq 2 \\ 2x+1 & \text{for } x > 2 \end{cases}$$

Evaluate  $f(-1) + 2f(2) - f(3)$ .

$$\begin{aligned}
 &= (-1)^2 + 2(2)^2 - (2(3)+1) \\
 &= 1 + 8 - 7 \\
 &= 2
 \end{aligned}$$

|   |         |
|---|---------|
| 2 | Correct |
| 1 | error   |

① if correct equation  
eg;  $x^2 + 2x^2 - (2x+1)$

- (c) If  $g(x) = x^2 + 3x + 5$ , show that  $g(a+1) = g(a) + 2(a+2)$ .

2

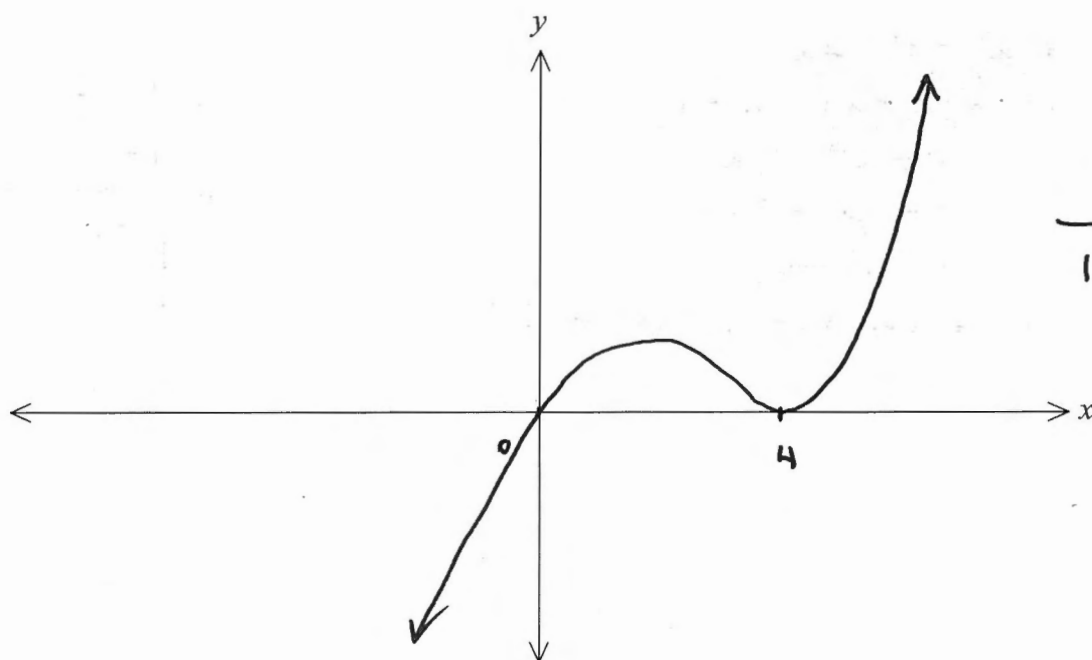
$$\begin{aligned}
 g(a+1) &= (a+1)^2 + 3(a+1) + 5 \\
 &= a^2 + 2a + 1 + 3a + 3 + 5 \\
 &= a^2 + 5a + 9 \\
 g(a) &= a^2 + 3a + 5 \\
 g(a) + 2(a+2) &= a^2 + 3a + 5 + 2a + 4 \\
 &= a^2 + 5a + 9 \\
 &= g(a+1) \text{ As req'd}
 \end{aligned}$$

|   |                                  |
|---|----------------------------------|
| 2 | Correct                          |
| 1 | Substituting $x = a+1$ correctly |

**Question 11 (6 marks)**

**Marks**

- (a) (i) On the axis provided, sketch the graph  $y = x(x-4)^2$ , showing all intercepts. 2



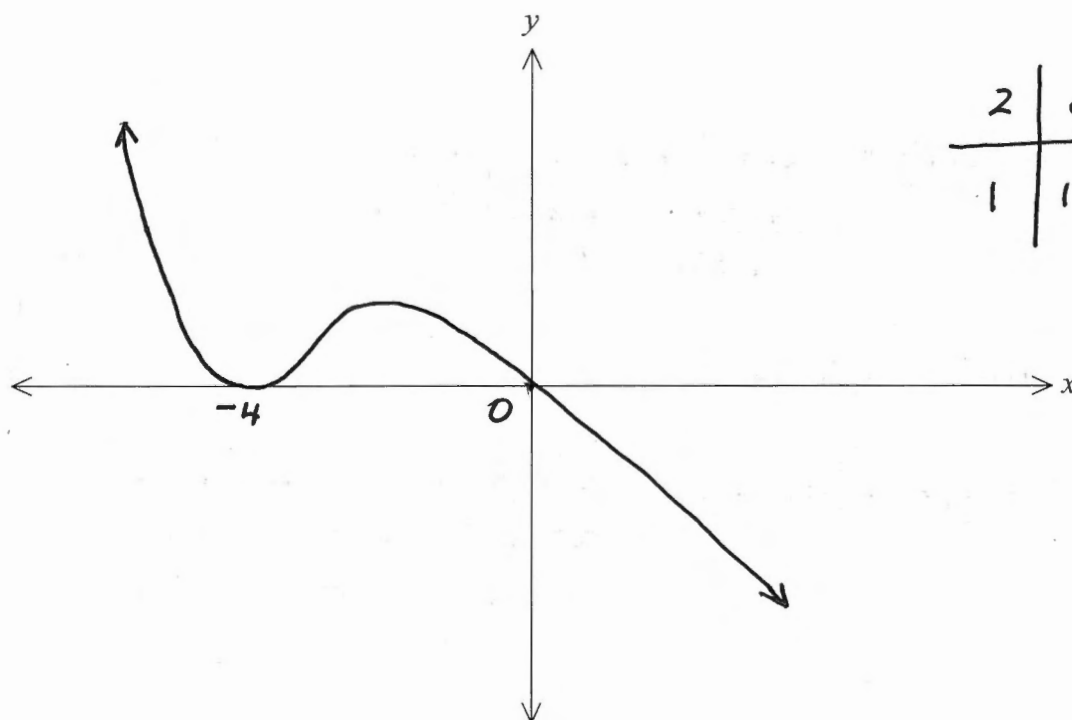
|   |         |
|---|---------|
| 2 | correct |
| 1 | error   |

- (ii) Hence, find the values of  $x$  for which  $x(x-4)^2 > 0$ . 2

$x \in (0, 4) \cup (4, \infty)$

|   |         |
|---|---------|
| 2 | correct |
| 1 | error   |

- (iii) On the number plane below, sketch  $y = f(-x)$  labelling all intercepts. 2



|   |         |
|---|---------|
| 2 | correct |
| 1 | error   |

Question 12 (7 marks)

Marks

- (a) Given that  $9x^3 + 3x^2 - 12x - 4 = (3x^2 - 4)(3x + 1)$ , hence, or otherwise, solve  $18x^4 + 6x^3 - 24x^2 - 8x = 0$ .

2

$$2x(9x^3 + 3x^2 - 12x - 4) = 0$$

$$2x(3x^2 - 4)(3x + 1) = 0 \quad *$$

$$\therefore x = 0 \text{ or } x^2 = \frac{4}{3} \text{ or } x = -\frac{1}{3}$$

$$x = \pm \frac{2}{\sqrt{3}}$$

|   |                                                             |
|---|-------------------------------------------------------------|
| 2 | Correct                                                     |
| 1 | Correct <span style="color: red;">✗</span><br>factorisation |

- (b) For what values of  $m$  does the equation  $6x^2 - 12mx - 7m + 3 = 0$  have equal roots.

3

$$\Delta = (-12m)^2 - 4(6)(3 - 7m)$$

$$= 144m^2 - 72 + 168m$$

$$= 24(6m^2 + 7m - 3)$$

|   |                                            |
|---|--------------------------------------------|
| 3 | Correct                                    |
| 2 | Correct $\Delta$<br>error in factorisation |
| 1 | $\Delta = 0$                               |

Equal roots  $\Delta = 0 \therefore 24(6m^2 + 7m - 3) = 0$

$$3m - 2 \quad 6m^2 + 7m - 3 = 0$$

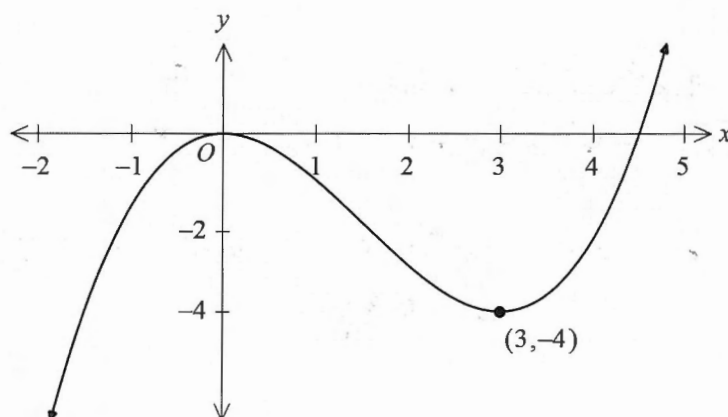
$$2m \times 3 \quad (3m - 2)(2m + 3) = 0$$

$$\therefore m = \frac{2}{3} \text{ or } m = -\frac{3}{2}$$

Question 12 continues on next page

Question 12 continued.

(c) The graph of  $y = f(x)$  is sketched below.



For what values of  $k$  does the equation  $f(x) = k$  have one solution?

2

$k > 0$  or  $k < -4$  ..... 2 | correct  
 .....  
 ..... 1 | one correct  
 .....  
 .....

Question 13 (4 marks)

Marks

Let  $f(x) = -x^2 + x + 4$  and  $g(x) = x^2 - 2$ .

(a) Find  $g(f(3))$ .

2

$$\begin{aligned}
 g(-3^2 + 3 + 4) &= g(-2) \\
 &= (-2)^2 - 2 \\
 &= 2
 \end{aligned}$$

|   |         |
|---|---------|
| 2 | correct |
| 1 | error   |

(b) Express  $f(g(x))$  in the form  $ax^4 + bx^2 + c$ , where  $a$ ,  $b$  and  $c$  are non-zero integers.

2

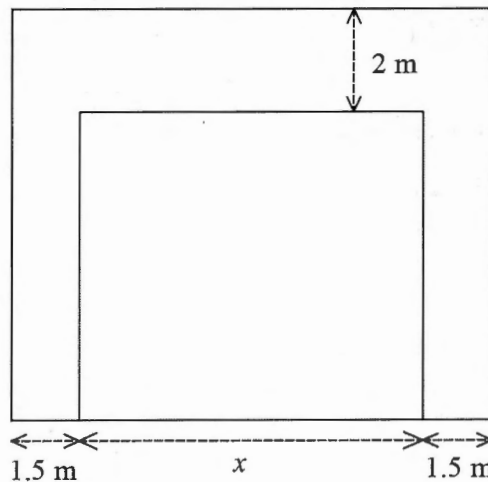
$$\begin{aligned}
 f(x^2 - 2) &= -(x^2 - 2)^2 + (x^2 - 2) + 4 * \\
 &= -(x^4 - 4x^2 + 4) + x^2 + 2 \\
 &= -x^4 + 5x^2 - 2
 \end{aligned}$$

|   |              |
|---|--------------|
| 2 | correct      |
| 1 | Correct to * |

**Question 14** (4 marks)

**Marks**

A square garage door of length  $x$  metres is surrounded on three sides, by brickwork. The brick work is 1.5 metres wide on each side of the door and has a height of 2 metres above the door, as shown below.



NOT TO SCALE

- (a) Find an expression for the area of the brickwork.

2

$$\begin{aligned} A_{\text{Brickwork}} &= (x+3)(x+2) - x^2 * \\ &= x^2 + 5x + 6 - x^2 \\ &= 5x + 6 \end{aligned}$$

|   |              |
|---|--------------|
| 2 | Correct      |
| 1 | Correct to * |

- (b) To meet building regulations, the area of the brick wall must be exactly twice the area of the garage door.

2

Find the value(s) of  $x$  that satisfy the building regulations.

$$\begin{aligned} 5x + 6 &= 2x^2 \\ 2x^2 - 5x - 6 &= 0 \end{aligned}$$

$$x = \frac{5 \pm \sqrt{25 + 48}}{4}$$

$$x = \frac{5 \pm \sqrt{73}}{4}$$

|   |                                                                                 |
|---|---------------------------------------------------------------------------------|
| 2 | Correct                                                                         |
| 1 | Correct quadratic and use of quadratic formula, not eliminating negative result |

But  $x$  is a length  $\therefore x > 0 \therefore x = \frac{5 + \sqrt{73}}{4}$  m.

**End of Assessment Task**